Quiz 2

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1. Consider the following IBVP:

$$\begin{aligned} u_{tt} - u_{xx} - u &= F(x,t), \qquad -\pi/2 < x < \pi/2, t > 1 \\ u_x(-\pi/2, t) = u_x(\pi/2, t) &= 0, \qquad t > 1 \\ u(x,1) = f(x), u_t(x,1) &= g(x), \qquad 0 < x < 2; \end{aligned}$$

where:

$$F(x,t) = 1 + 2\sin(x/2), f(x) = 0, g(x) = \sin(x).$$

Proof (Hernandez). We seek:

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t)\varphi_n(x),\ F(x,t) = \sum_{n=1}^{\infty} \beta_n\varphi_n(x),\ g(x) = \sum_{n=1}^{\infty} \alpha_n\varphi_n(x),$$

where the eigenfunctions are given by:

$$-\varphi_n{}'' - \varphi_n - \lambda_n^2\varphi = -\varphi_n{}'' - (1+\lambda_n^2)\varphi_n = 0, \qquad \varphi'(-\pi/2) = \varphi'(\pi/2) = 0,$$

whose general solution is:

$$\begin{aligned} \varphi_n(x) &= c_1 e^{x\sqrt{-\lambda_n^2 - 1}} + c_2 e^{-x\sqrt{-\lambda_n^2-1}} = e^{0x}(A\cos(x(\lambda_n^2+1)) + B\sin(x(\lambda_n^2+1))) \\ \varphi_n{}'(x) &= B\cos(x(\lambda_n^2+1))(\lambda_n^2+1) - A\sin(x(\lambda_n^2+1))(\lambda_n^2+1). \end{aligned}$$

We obtain the particular solution by applying the IBV conditions and computing the Fourier coefficients.

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